

Effects of Protein Supplements on Muscle Damage, Soreness and Recovery of Muscle Function and Physical Performance: A Systematic Review

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Abstract

Background Protein supplements are frequently consumed by athletes and recreationally-active individuals, although the decision to purchase and consume protein supplements is often based on marketing claims rather than evidence-based research.

Objective To provide a systematic and comprehensive analysis of literature examining the hypothesis that protein supplements enhance recovery of muscle function and physical performance by attenuating muscle damage and soreness following a previous bout of exercise.

Data Sources English language articles were searched with PubMed and Google Scholar using protein and supplements together with performance, exercise, competition and muscle, alone or in combination as keywords.

Study Selection Inclusion criteria required studies to recruit healthy adults less than 50 years of age and to evaluate the effects of protein supplements alone or in combination with carbohydrate on performance metrics including time-to-exhaustion, time-trial or isometric or isokinetic muscle strength and markers of muscle damage and soreness. Twenty-seven articles were identified of which 18 dealt exclusively with ingestion of protein supplements to reduce muscle damage and soreness and improve recovery of muscle function following exercise, whereas the remaining 9 articles assessed muscle damage

as well as performance metrics during single or repeat bouts of exercise.

Study Appraisal and Synthesis Methods Papers were evaluated based on experimental design and examined for confounders that explain discrepancies between studies such as dietary control, training state of participants, sample size, direct or surrogate measures of muscle damage, and sensitivity of the performance metric.

Results High quality and consistent data demonstrated there is no apparent relationship between recovery of muscle function and ratings of muscle soreness and surrogate markers of muscle damage when protein supplements are consumed prior to, during or after a bout of endurance or resistance exercise. There also appears to be insufficient experimental data demonstrating ingestion of a protein supplement following a bout of exercise attenuates muscle soreness and/or lowers markers of muscle damage. However, beneficial effects such as reduced muscle soreness and markers of muscle damage become more evident when supplemental protein is consumed after daily training sessions. Furthermore, the data suggest potential ergogenic effects associated with protein supplementation are greatest if participants are in negative nitrogen and/or energy balance.

Limitations Small sample numbers and lack of dietary control limited the effectiveness of several investigations. In addition, studies did not measure the effects of protein supplementation on direct indices of muscle damage such as myofibrillar disruption and various measures of protein signaling indicative of a change in rates of protein synthesis and degradation. As a result, the interpretation of the data was often limited.

Conclusions Overwhelmingly, studies have consistently demonstrated the acute benefits of protein supplementation on post-exercise muscle anabolism, which, in theory, may

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facilitate the recovery of muscle function and performance. However, to date, when protein supplements are provided, acute changes in post-exercise protein synthesis and anabolic intracellular signaling have not resulted in measurable reductions in muscle damage and enhanced recovery of muscle function. Limitations in study designs together with the large variability in surrogate markers of muscle damage reduced the strength of the evidence-base.

1 Introduction

Protein supplements are one of the most popular dietary supplements used by athletes, recreationally active adults and soldiers [1–4]. When surveyed, individuals commonly cite expectations for increased muscle mass, improved exercise recovery and improved performance [2, 5] as reasons for protein supplement use and respondents typically indicate they rely on coaches, teammates and family or friends to gain information about these products [1, 5]. Marketing claims have convinced many consumers that protein supplements, whether consumed alone or in combination with carbohydrate, will enhance physical performance by attenuating carbohydrate oxidation during prolonged steady-state exercise and will hasten muscle glycogen repletion during recovery. In particular, consumers also believe protein supplementation will promote lean muscle mass accretion by enhancing rates of protein synthesis and decreasing rates of protein degradation, and optimize recovery of muscle function and physical performance following exercise by attenuating muscle damage and muscle soreness.

Acute unaccustomed exercise that includes heavy eccentric loading such as resistance training or downhill running can damage contractile proteins, inducing muscle soreness, which can last for several days, and may impair muscle function (for review see Clarkson et al. [6]). In these circumstances, rates of muscle protein synthesis and degradation are increased [7–9], but in the absence of adequate nutrition rates of degradation exceed synthesis resulting in a negative net protein balance [10–14]. However, consuming protein supplements during recovery from exercise promotes skeletal muscle anabolism [15, 16], stimulating greater rates of myofibrillar and mitochondrial protein synthesis [17, 18]. In theory, the stimulation of muscle protein synthesis by protein supplementation represents a critical skeletal muscle adaptive response to mechanical stress that aids in the growth and repair of contractile proteins, thereby facilitating long-term recovery by promoting muscle remodeling [10, 18–21]. As proposed by Saunders [22], these changes in protein synthesis with supplementation should reduce indices of muscle damage and hasten the recovery of muscle function. However, a systematic review of the evidence to support or refute the

relationship between the use of protein supplements before, during or following exercise and reductions in markers of muscle damage and the recovery of muscle function does not exist. As such, this paper systematically reviews studies that examine indices of muscle damage and recovery of muscle function and physical performance when protein supplements are provided alone or in combination with carbohydrate for several hours or days after a bout of resistance or endurance exercise. The review also includes a strength of recommendation [23] judgment for the expected performance benefits associated with the recovery of muscle function after consuming protein supplements.

Due to the volume of material retrieved, separate reviews will address the evidence-base support for the use of protein supplements alone or in combination with carbohydrate to enhance physical performance during acute or repeated bouts of endurance exercise, as well as to hasten the accretion of muscle mass and strength.

2 Methods

Literature searches not restricted by publication date were conducted with PubMed and Google Scholar using keywords that included protein and supplements together with performance, exercise, competition and muscle, alone or in combination. Only English language articles were retrieved. Searches were restricted to peer-reviewed publications reporting findings from healthy human adults between 18 and 50 years of age who were consuming dietary protein at or above the recommended dietary allowance of $0.8 \text{ g} \cdot \text{kg}^{-1} \cdot \text{day}^{-1}$ as part of their normal habitual diet [24]. Articles that involved dietary manipulations to compare the effects of protein or carbohydrate on performance were not included [25, 26]. Papers that tested the efficacy of protein supplementation on indirect markers of muscle damage following exercise but did not include a subsequent test of muscle function or physical performance were also excluded [27–30]. Studies that compared drink formulations that included protein as well as vitamins and antioxidant supplements were not included since it was not possible to isolate the effects attributable to protein content alone [31]. Only studies that reported findings with the ingestion of various forms of protein alone or in combination with carbohydrate were reviewed. Articles were also retrieved from the reference lists of these papers and recent reviews on protein supplements. Papers were examined in detail, searching for potentially confounding experimental design issues that could explain discrepant findings observed across studies, such as the energy content of the supplements, dietary control, use of trained or untrained participants, number of participants tested and sensitivity of the test metrics.

3 Results

Our search identified 27 articles (Fig. 1) of which 18 dealt exclusively with the ingestion of protein supplements to reduce muscle damage and soreness and improve the recovery of muscle function and physical performance following exercise, whereas the remaining 9 articles assessed measures of muscle damage, as well as performance metrics, during single or repeated bouts of exercise. A strength of recommendation taxonomy (SORT) [23] was used to establish quality of evidence for conclusions specific for each of these categories. The SORT uses the following criteria: A, recommendation based on consistent and good-quality experimental evidence; B, recommendation based on inconsistent or limited-quality experimental evidence; or C, recommendation based on consensus, opinion, usual practice, case studies or extrapolation from quasi-experimental research. The SORT was created to assist medical practitioners in their assessment of patient-oriented evidence [23]. In a similar manner, SORT can be used to assess the evidence-based literature to provide recommendations of protein supplement use by athletes and recreationally active adults.

In the context of this review, metrics for physical performance involved cycling or running tests-to-exhaustion at a given exercise intensity or time-trials over a set distance. Tests of muscle function included isometric, isokinetic or dynamic measures of muscle strength (such as 1

repetition maximum) or muscular endurance involving the number of repetitions performed over a defined range of motion and resistance. Serum or plasma creatine kinase (CK) [6] and urinary 3-methylhistidine [32] concentrations were often assessed as indirect markers of muscle protein damage.

3.1 Tests of Physical Performance, Muscle Function and Damage Within 24 h

The use of protein supplements may be beneficial for athletes and recreationally active adults who engage in exercise training on a daily basis, and often train or compete more than once per day. To test this theory, several studies have assessed whether protein supplements reduce the extent of muscle damage and improve subsequent muscle function and physical performance during a 24-h period after an initial bout of exercise. The presentation of the findings for this section is divided into subsections categorized based on mode of exercise-induced muscle damage. The outcome measures from these studies are summarized in Table 1.

3.1.1 Studies Conducted Following Submaximal Cycling

Saunders et al. [33] studied the effects of carbohydrate and protein supplementation during repeated bouts of exercise performance separated by several hours. Their

Fig. 1 Study selection and flow diagram of articles included in the review

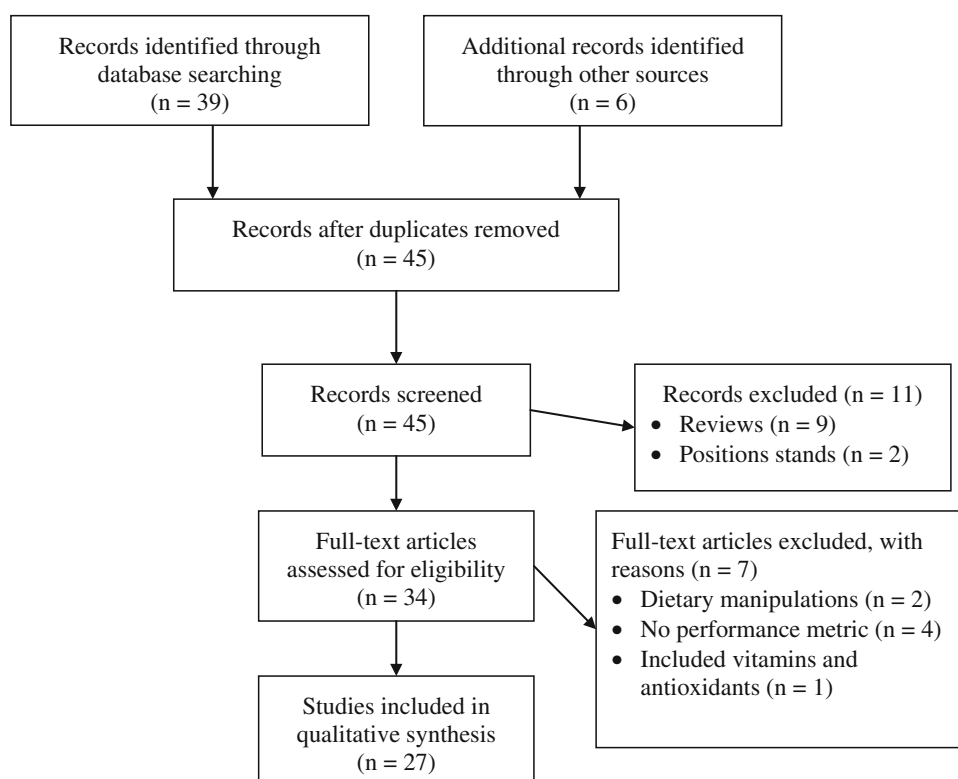


Table 1 A summary of the studies that followed changes in muscle damage, soreness and recovery of muscle function for 24 h or less with protein and/or carbohydrate supplementation following an initial bout of acute exercise

Study and participants ^a	Initial type of exercise	Supplementation dose and timing	Muscle damage (CK, LDH, Mb)	Muscle soreness	Muscle function
Breen et al. [36], TR (12)	Cycling	6 % CHO 6 % CHO + 1.8 % PRO 270 mL/15 min during 2 h exercise	CHO + PRO = CHO	CHO + PRO = CHO	CHO + PRO = CHO for IMST
Buckley et al. [42], UT 3 groups (28)	Eccentric leg	10 % PRO PLA 250 mL immediately following and 2 and 22 h post exercise	PRO = PLA	PRO = PLA	PRO > PLA for IMST
Cermak et al. [35], TR (8)	Cycling	6 % CHO 6 % CHO + 2 % PRO 250 mL/15 min during 90 min exercise	CHO + PRO = CHO		CHO + PRO = CHO for TT
Ferguson-Stegall et al. [38], TR (10)	Cycling	1.2 g/kg CHO M (0.9 g/kg CHO + 0.3 g/kg PRO) PLA 16 mL/kg over 4 h recovery	M = CHO = PLA		M < CHO = PLA for TT
Millard-Stafford et al. [41], TR (8) and TR (24)	Running	10.3 % CHO 8 % CHO + 2.3 % PRO 6 % CHO 1 g/kg/h over first 2 h recovery plus 700 mL later	CHO + PRO < CHO	CHO + PRO < CHO	CHO + PRO = CHO for TTE and TT
Pritchett et al. [40], TR (10)	Cycling	1 g/kg CHO + 0.25 g/kg PRO CM (1 g/kg CHO + 0.25 g/kg PRO) Each hour for 2 h recovery	M < CHO + PRO	M = CHO + PRO	M = CHO + PRO for TTE
Romano-Ely et al. [34], TR (14)	Cycling	18.6 % CHO 15 % CHO + 3.6 % PRO 10 mL/kg post exercise	CHO + PRO < CHO	CHO + PRO < CHO	CHO + PRO = CHO for TTE
Rowlands et al. [39], TR (10)	Cycling	2.35 g/kg/h CHO 1.6 g/kg/h CHO + 0.8 g/kg/h PRO Over 4 h of recovery	CHO + PRO < CHO	CHO + PRO < CHO	CHO + PRO = CHO for sprints
Saunders et al. [33], TR (15)	Cycling	7.3 % CHO 7.3 % CHO + 1.8 % PRO 10 mL/kg post exercise	CHO + PRO < CHO		CHO + PRO > CHO for TTE
Valentine et al. [37], TR (11)	Cycling	7.8 % CHO 9.7 % CHO 7.8 % CHO + 1.9 % PRO PLA 250 mL/15 min during exercise	CHO + PRO < [CHO = PLA]	CHO + PRO = CHO = PLA	CHO + PRO > CHO > PLA for reps at 70 % 1 RM

CHO carbohydrate, CK creatine kinase, CM chocolate milk, IMST isometric strength, LDH lactate dehydrogenase, M milk, Mb myoglobin, PLA placebo, PRO protein, RM repetition maximum, TR trained, TT time-trial, TTE time-to-exhaustion, UT untrained

^a Participant numbers are provided in parentheses

experimental design included providing participants with a bolus dose of these supplements during the 30 min that followed their first cycling test-to-exhaustion at 75 % of maximal aerobic power ($\dot{V}O_{2\max}$) and then comparing cycling time-to-exhaustion at 85 % $\dot{V}O_{2\max}$ 12–15 h later. Plasma CK levels were reduced during the recovery period and subsequent times-to-exhaustion were increased by 40 % with carbohydrate + protein supplementation compared with carbohydrate only. However, the supplements were not isocaloric. The carbohydrate + protein beverage provided 25 % more energy.

The effects of matching the caloric content of the protein and/or carbohydrate supplements ingested during and following exercise have also been studied. Romano-Ely et al. [34] provided a bolus isocaloric carbohydrate or carbohydrate + protein beverage immediately following an initial cycling test-to-exhaustion at 70 % $\dot{V}O_{2\max}$ and then measured indices of muscle damage and soreness during recovery prior to a subsequent exercise test-to-exhaustion 24 h later at 80 % $\dot{V}O_{2\max}$. Despite increased plasma CK levels and ratings of muscle soreness during recovery, with carbohydrate times-to-exhaustion at 80 % $\dot{V}O_{2\max}$ were approximately 43 min in duration and similar across conditions. Others have confirmed the lack of a beneficial effect of combined carbohydrate + protein supplementation on subsequent muscle performance when compared to an isocaloric control. Cermak et al. [35] reported that carbohydrate alone or with additional protein provided during 90 min of cycling at approximately 70 % $\dot{V}O_{2\max}$ resulted in similar increases in plasma CK levels and 20-km time-trial performance the following day. Similarly, Breen et al. [36] showed that consumption of carbohydrate alone or with additional protein during prolonged cycling had no effect on the decrease in isometric knee extensor strength or the increases in muscle soreness or plasma CK levels recorded 24 h later.

In contrast, some reports have demonstrated a potential ergogenic effect of combined carbohydrate and protein supplementation. For example, Valentine et al. [37] reported reduced plasma CK levels and increased repetitions at 70 % leg extension maximum 24 h following the completion of a cycling test-to-exhaustion at 75 % $\dot{V}O_{2\max}$ that was accompanied with ingestion of an isocaloric carbohydrate + protein beverage. In addition, Ferguson-Stegall et al. [38] examined the effects of isocaloric carbohydrate or low-fat chocolate milk supplementation during a 4-h recovery period that followed a 100-min glycogen depleting exercise bout on biomarkers of muscle damage and inflammation, and subsequent 40-km time-trial performance. In that study, consuming the isocaloric low-fat chocolate milk, a combination of carbohydrate + protein, had no effect on CK levels

and circulating cytokine markers of inflammation when compared to the carbohydrate control. However, time-trial performance was approximately 6 % faster following recovery supplementation with chocolate milk.

3.1.2 Studies Conducted Following High-Intensity Interval Cycling

Rowlands et al. [39] evaluated the efficacy of a protein enriched recovery supplement on sprint cycling performance 15 h following 2.5 h of high-intensity interval cycling that simulated a 100-km road race. Immediately following and at 30-min intervals over a 4-h recovery period, participants consumed isocaloric carbohydrate + protein supplements with low or high protein content. The protein enriched supplement reduced plasma CK levels and leg soreness, but total power output during ten successive sprints as well as the expected decline in power were not different between the high and low protein supplements. Similarly, Pritchett et al. [40], who compared the effects of low-fat chocolate milk or an isocaloric carbohydrate + protein supplement consumed during a 2-h recovery period on time-to-exhaustion at 85 % $\dot{V}O_{2\max}$ approximately 15 h following 50 min of high-intensity interval cycling, failed to observe an ergogenic effect, as performance times were not different between treatments independent of plasma CK.

3.1.3 Studies Conducted Following Submaximal Running

Millard-Stafford et al. [41] conducted two studies of the impact of the caloric content of a carbohydrate and carbohydrate + protein supplement on recovery of performance both during the same day and the day after a 21-km training run and an initial run-to-exhaustion at 90 % $\dot{V}O_{2\max}$. In one study, a repeated measures design was used to test the efficacy of carbohydrate or carbohydrate + protein beverages provided during a 2-h recovery period following the initial run-to-exhaustion. An additional 700 mL of supplement was consumed prior to a 5-km time-trial the following day. Since the study was also designed to assess the impact of protein supplementation on muscle soreness, a second experiment was conducted that used a between group design with runners matched on their season's performance and then randomly allocated to receive one of the beverages, thereby avoiding the known repeated-bout effect on muscle soreness [6]. For both studies, ratings of muscle soreness were reduced with carbohydrate + protein but plasma CK levels and recovery of performance during the same day or the next day were similar regardless of supplement consumed.

3.1.4 Studies Conducted Following Eccentric Resistance Exercise

Buckley et al. [42] induced muscle damage in untrained volunteers who consumed either placebo, whey protein or a fast-absorbing hydrolyzed whey protein supplement immediately after completing 100 eccentric knee extension repetitions and again at 6 and 22 h during recovery. Isometric force returned to baseline values within 6 h for those receiving the hydrolyzed whey protein, which was significantly faster than those receiving placebo or whey protein. Interestingly, isometric force was similar and remained lower compared with baseline for the duration of the 24-h recovery period for volunteers consuming placebo and whey protein. Although consuming hydrolyzed whey protein improved muscle function recovery, subjective ratings of muscle soreness and circulating CK concentrations were not different over the 24-h recovery period.

3.1.5 Summary: Tests of Muscle Function and Physical Performance Within 24 h

Some studies have reported reduced muscle soreness and/or CK levels after consuming a combined carbohydrate + protein supplement, yet find no effect on subsequent time-to-exhaustion [34, 40, 41] or repeated sprint intervals performed within 24 h of the initial endurance exercise bout [39]. However, others have reported reductions in CK levels consequent to carbohydrate + protein supplementation, which were associated with concomitant improvements in muscle force during repeat muscle contractions at 70 % 1-repetition maximum [37] or cycling time-to-exhaustion [33]. Still others reported improvements in subsequent endurance performance with the ingestion of carbohydrate + protein despite observing no changes in CK levels in response to prolonged submaximal cycling [38]. Finally, there were also reports of no change in CK following cycling exercise and no effect on subsequent 20-km time-trial performance [35] or isometric knee extensor strength [36] with a carbohydrate + protein supplement. Regardless of whether carbohydrate + protein supplementation occurred during, immediately following or several hours into recovery, and regardless of the type and dose of protein supplementation, the findings of these studies have been inconsistent, and suggest minimal relationships between subjective ratings of muscle soreness and indirect biomarkers of muscle damage and subsequent tests (i.e., within 24 h) of muscle function or physical performance [34–36, 38–41]. The lone study that used an eccentric exercise model did not observe a relationship between recovery of muscle function with protein supplementation and markers of muscle damage but did report that muscle function recovered more quickly with hydrolyzed whey protein supplementation [42].

3.2 Tests of Muscle Function and Damage over Several Days

Following a bout of novel eccentric or sustained endurance-type exercise the recovery of isometric and/or isokinetic muscle strength may require several days to return to baseline values, consistent with the time course for remodeling of contractile protein [9]. Therefore, studies have examined the effects of protein supplementation alone or in combination with carbohydrate on the recovery of muscle function for periods longer than 24 h. To organize the presentation of these findings, this section is divided into four subsections categorized based on mode of exercise-induced muscle damage. The outcome measures from these studies are summarized in Table 2.

3.2.1 Studies Conducted Following Submaximal Cycling

Given the role of branched-chain amino acids (BCAA) in skeletal muscle protein metabolism during steady-state exercise, and the potent stimulatory effects of post-exercise BCAA administration on measures of whole-body and muscle protein synthesis, Greer et al. [43] compared effects of isocaloric carbohydrate or carbohydrate + BCAA supplementation immediately before and after 1 h of a 90-min bout of submaximal exercise at 55 % $\dot{V}O_{2\max}$ on indices of muscle damage and recovery in physically inactive adults. Markers of muscle damage were attenuated with the carbohydrate + BCAA and carbohydrate supplements compared with PLA throughout a 48-h recovery period. Plasma CK levels were actually lower for carbohydrate + BCAA 24 h after exercise compared to carbohydrate and ratings of muscle soreness were also lower at 24 h following carbohydrate + BCAA supplementation as compared to the other treatments. Although peak torque was reduced similarly between treatments, the recovery of leg flexion torque was faster with the carbohydrate + BCAA, as values 48 h post-exercise were greater for carbohydrate + BCAA as compared with other treatments. These data suggest addition of BCAA to a carbohydrate beverage consumed prior to and after exercise will attenuate muscle damage and hasten the recovery of muscle function.

3.2.2 Studies Conducted Following Downhill or Shuttle Running

Since the effects of unaccustomed eccentric exercise on muscle damage, particularly in untrained individuals, are well established, studies have assessed whether protein supplementation enhances recovery to eccentric exercise-induced muscle damage. Green et al. [44] measured muscle soreness, plasma CK and maximal voluntary contraction (MVC) immediately following, and for 3 days after,

Table 2 A summary of the studies that followed changes in muscle damage, soreness and recovery of muscle function for several days with protein and/or carbohydrate supplementation following an initial bout of cycling, running, marching or eccentric resistance exercise

Study and participants ^a	Initial type of exercise	Supplementation dose and timing	Muscle damage (CK, LDH, Mb)	Muscle soreness	Muscle function
Betts et al. [47], TR (17)	Shuttle running	9 % CHO 9 % CHO + 3 % PRO 7 mL/kg before, 5 × 2.6 mL/kg during and 8 × 6.7 mL/kg after exercise 6.4 % CHO 7 % PRO PLA 500 mL during exercise and 2 × 500 mL after testing for 3 days	CHO + PRO = CHO	CHO + PRO = CHO	CHO + PRO = CHO for knee and hip flexion/extension
Blacker et al. [48], TR (10)	Load carriage and marching	6.4 % CHO 7 % PRO PLA			PRO = CHO > PLA for leg IMST PRO = CHO = PLA for leg, trunk IKST and LFF
Cockburn et al. [50], TR 4 groups (24)	Eccentric leg	CM (118 g CHO + 33 g PRO) M (49 g CHO + 34 g PRO) 64 g CHO PLA 500 mL immediately after and 2 h following exercise	[CM = M] < [CHO = PLA]	CM = M = CHO = PLA	[CM = M] > [CHO = PLA] for leg IKST and TW
Etheridge et al. [45], TR (9)	Downhill running	100 g PRO PLA Immediately after exercise	PRO = PLA	PRO = PLA	PRO > PLA for IMST and PPO
Green et al. [44], TR 3 groups (18)	Downhill running	0.6 g/kg CHO 0.6 g/kg CHO + 0.15 g/kg PRO PLA Immediately following and 30 and 60 min after exercise	CHO + PRO = CHO = PLA	CHO + PRO = CHO = PLA	CHO + PRO = CHO = PLA for IMST
Greer et al. [43], UT (9)	Cycling	6 % CHO 5.4 % CHO + 0.6 % PRO PLA 400 mL before and during exercise	CHO + PRO < CHO < PLA	CHO + PRO < [CHO = PLA]	CHO + PRO = CHO = PLA for leg extension CHO + PRO > CHO = PLA for leg flexion
Hoffman et al. [55], TR 2 groups (15)	Resistance training	42 g PRO PLA Before and after exercise for 3 days	PRO = PLA	PRO = PLA	PRO > PLA for TW
Jackman et al. [51], UT 2 groups (24)	Eccentric leg	29.2 g/day PRO PLA Four equal servings between meals daily for 3 days	PRO = PLA	PRO < PLA	PRO = PLA for knee ISMT and LFF
Nosaka et al. [52], UT (38)	Eccentric arm	7.2 g/day PRO PLA Equal servings twice daily for 3 days	PRO < PLA	PRO < PLA	PRO = PLA for arm IMST

Table 2 continued

Study and participants ^a	Initial type of exercise	Supplementation dose and timing	Muscle damage (CK, LDH, Mb)	Muscle soreness	Muscle function
White et al. [53], UT 3 groups (27)	Eccentric leg	23 g PRO + 75 g CHO PLA Immediately before or after exercise	CHO + PRO = PLA	CHO + PRO = PLA	CHO + PRO = PLA for knee IMST
Wojcik et al. [54], UT 3 groups (26)	Eccentric leg	1.3 g/kg CHO M (0.9 g/kg CHO + 0.4 g/kg PRO) PLA Immediately after and 2 h following exercise	M = CHO = PLA	M = CHO = PLA	M = CHO = PLA for leg IKST and TW

CHO carbohydrate, CK creatine kinase, CM chocolate milk, IKST isokinetic strength, IMST isometric strength, LFF low frequency fatigue, M milk, Mb myoglobin, PLA placebo, PRO peak power output, PRO protein, TR trained, TW total work, UT untrained

^a Participant numbers are provided in parentheses

30 min of downhill treadmill running. Participants received non-isocaloric placebo, carbohydrate alone or with additional protein immediately after and again 30 and 60 min following the exercise. For all groups MVC was 20 % lower immediately after exercise and returned to baseline levels similarly for each group after 3 days of recovery. Ratings of muscle soreness and plasma CK levels were also affected equally across groups and the pattern of change was similar during the recovery period. Thus, consuming multiple doses of a carbohydrate beverage with an additional 30 g of protein during the initial 60 min after completing a bout of novel downhill running does not appear to alter muscle damage and function.

Etheridge et al. [45] examined the effects of a larger 100 g dose of protein consumed immediately following a 30-min downhill treadmill running session on recovery of MVC, peak power output, muscle soreness and CK. Protein had no effect on changes in muscle soreness and CK levels over the subsequent 72 h. However, in contrast to Green et al. [44], recovery of muscle function was enhanced with protein supplementation, as protein ingestion prevented declines in MVC observed with placebo. Thus, the 100 g of protein consumed improved muscle function following downhill running, a beneficial effect not related to ratings of muscle soreness and/or markers of muscle damage.

Protein supplementation may also be efficacious in response to high-intensity intermittent shuttle running, which can induce muscle injury due to frequent changes in direction and decelerations from repeated sprints [46]. Betts et al. [47] evaluated whether repeated protein supplementation totaling 160 g during and after a demanding 90-min bout of shuttle running influenced muscle function, as indicated by hip and knee flexion and extension force, as well as indices of muscle damage. Isometric torque was decreased following shuttle running by approximately 20 %. However, 7-day recovery of muscle function was not different between carbohydrate and carbohydrate + protein treatments, which were isocarbohydrate but not isocaloric. Similarly, neither increases in muscle soreness nor plasma CK levels were different between treatments. Thus, these findings are consistent with Green et al. [44] yet in contrast to Etheridge et al. [45], and indicate that additional protein supplementation, regardless of dose, with an optimal delivery rate of carbohydrate does not improve the recovery of muscle function or reduce surrogate markers of muscle injury following unaccustomed downhill or shuttle running.

3.2.3 Studies Conducted Following Prolonged Load Carriage

Load carriage, a mode of exercise typically performed by military personnel for sustained durations, produces

mechanical strain and metabolic demands on skeletal muscle that are unlike those seen with common endurance- and resistance-type exercise modalities. To assess whether protein supplementation prevents decrements in muscle performance following a 2-h load carriage (25-kg backpack) exercise bout, Blacker et al. [48] evaluated a non-isocaloric placebo, carbohydrate or protein supplement consumed prior to, immediately following and again after each subsequent testing session during 3 days of recovery. Isometric strength decreased similarly across conditions ($\sim 15\%$), but returned more rapidly to baseline levels after 48 h for carbohydrate or protein compared with placebo. Changes in low frequency fatigue, a measure commonly used to document muscle damage [49], and isokinetic strength of the knee, trunk or shoulder during recovery were similar among treatments. Thus, although protein or carbohydrate supplementation promoted faster recovery of isometric strength, neither supplement was associated with improved recovery of isokinetic muscle function or measures of fatigue.

3.2.4 Studies Conducted Following Eccentric Resistance Exercise

Results of the studies that examined effects of protein supplementation on indices of muscle damage and subsequent measures of function in response to resistance eccentric exercise with naïve participants have been inconsistent [50–54]. For example, ingesting carbohydrate + protein either immediately before or after eccentric resistance exercise failed to influence ratings of muscle soreness, plasma CK, and recovery measures of muscle strength [53]. Similarly, consuming supplemental protein (skim milk) immediately and 2 h after eccentric leg exercise failed to affect circulating markers of muscle damage and inflammation (e.g., CK, interleukin-1, interleukin-6, and tumor necrosis factor alpha), muscle soreness, and recovery measures of peak torque and total work when compared to placebo [54]. Similarly, Nosaka et al. [52] failed to demonstrate an acute benefit (i.e., improvements in strength, range of motion, soreness, indices of damage) of consuming a 60 % essential amino acid (EAA) solution prior to and immediately after a 30-min bout of elbow flexion and extension eccentric exercise, further suggesting that protein supplementation does not promote faster recovery of muscle function or reduction in muscle soreness after eccentric exercise. However, when EAA supplementation was continued over 4 days of recovery, ratings of muscle soreness and circulating indices of muscle damage were reduced compared with placebo, suggesting that continued provision of EAA may contribute to long-term recovery from eccentric exercise. Results

from Jackman et al. [51], in part, support this theory, as repeat BCAA supplementation (7.3 g) prior to and four times daily for 72 h into recovery from eccentric exercise attenuated muscle soreness compared to placebo in untrained individuals, findings that are similar to earlier work [30]. However, BCAA supplementation had no effect on measures of muscle function and damage [51], suggesting BCAA alone may not be sufficient to promote recovery from eccentric resistance exercise.

Consistent with these positive findings, Cockburn et al. [50], who administered either placebo, carbohydrate, or a low-fat chocolate milkshake or low-fat milk supplement (high-quality sources of EAA) following eccentric knee flexion exercise, demonstrated that despite similar increases in muscle soreness over 48 h, impairment in knee extension torque, total work performed during six contractions and changes in plasma CK were lower for those consuming milk-containing products compared to carbohydrate alone or placebo. These findings also suggest amino acid availability, particularly EAA, may attenuate muscle damage and enhance muscle recovery. However, it is important to recognize that although protein content of the milkshake and milk supplements was equal, total caloric content of the beverages was not controlled as the milkshake and milk products contained 150 and 70 % more calories than the carbohydrate beverage, respectively. Nevertheless, since protein content was similar between the milk-based supplements, but carbohydrate content was different, these findings support the view consuming protein immediately after eccentric resistance exercise will attenuate indirect markers of muscle damage and improve muscle function.

The effects of protein supplementation immediately before and after a bout of resistance exercise on markers of muscle damage and recovery of muscle function in resistance trained individuals were studied by Hoffman et al. [55]. Volunteers consumed either placebo or a protein supplement before and after completing four sets of resistance exercises with follow-on performance measures performed 24 and 48 h later. Muscle performance, as indicated by the number of successful repetitions performed on a predetermined exercise session 24 and 48 h after the initial training session, was higher for athletes consuming protein compared with placebo. However, there was no difference in ratings of muscle soreness or the ~ 2 –3 fold increase in plasma CK levels. To the best of our knowledge, this is the first study demonstrating a benefit of protein supplementation, consumed immediately before and after resistance-training, on muscle function recovery in trained volunteers. However, since all participants were regularly engaged in resistance-training, the experimental design would have been strengthened by including a repeated-measures rather than an independent group comparison.

3.2.5 Summary: Tests of Muscle Function over Several Days

The most consistent finding from studies evaluating recovery of muscle function over several days for untrained participants in response to an acute bout of exercise is that protein supplementation alone or when combined with carbohydrate does not confer a metabolic advantage that hastens muscle recovery [43, 44, 47, 48, 51–54]. This was true regardless of whether muscle function was evaluated at 24 h, in agreement with Sect. 3.1.5, or longer into the recovery period. However, subjective ratings of muscle soreness are often reduced with protein ingestion (Table 2) [43, 50–52]. It is also important to note that dose and type of protein (BCAA, EAA, whey isolate or milk) consumed in these studies varied, thereby complicating the generalizations that can be made. In addition, one investigation demonstrating faster recovery of muscle function with the ingestion of protein supplements did not include dietary monitoring during the recovery period [50], a limitation that weakens the internal validity of the supplementation protocol. Interestingly, the lone study that recruited elite power lifters demonstrated improved recovery of muscle function with protein supplementation before and after a single training session [55] but these unique findings need to be replicated in other laboratories.

3.3 Multiple Days of Exercise and Supplementation

Among both athletes and adults who are physically active on a regular basis or exercise frequently to maintain or improve fitness, use of protein supplements is often considered a reliable option to hasten recovery and reduce muscle soreness before the next training session. In this section, the six studies examining the effects of protein supplementation for several days to weeks during repeated exercise sessions on markers of muscle damage and/or muscle function and physical performance are reviewed (see Table 3 for summary). These studies are presented in three subsections based on the nature of the exercise challenge.

3.3.1 A Study of Recruit Training

Flakoll et al. [56] were the first to assess long-term effects of protein supplementation on health, muscle soreness and function during military basic combat training. US Marine recruits were randomly assigned to placebo, carbohydrate or carbohydrate + protein (10 g protein). Nearly 130 participants were assigned to each group. Supplements were provided following each training session, which were conducted every other day of the 54-day program. All recruits had access to the same food and consumed meals at scheduled times in military dining facilities, although

dietary intake was not measured or controlled. Ratings of muscle soreness were significantly reduced for the carbohydrate + protein group following the mid-program 9.6-km march and final fitness testing. In addition, total medical visits as well as treatments for infections, muscle and joint problems or heat exhaustion were all reduced for the carbohydrate + protein group. Despite the small dose of protein consumed, the authors suggested timing of the ingestion immediately post exercise demonstrated the benefit of ingesting small amounts of protein for increasing protein accretion [14], thereby reducing muscle damage and soreness after exercise. The substantial reduction in medical visits certainly warrants further study. The authors also stated it was quite unlikely that the additional 40 kcal from the protein would account for the differences observed between the carbohydrate and carbohydrate + protein groups, especially considering that differences in caloric content between the carbohydrate and placebo supplements was much larger but there were no differences in any of the measures between these groups.

3.3.2 Studies of Cycle Training and Road Races

Skillen et al. [57] examined the effects of an isocaloric carbohydrate or carbohydrate + BCAA supplement provided prior to, during and after repeated testing and training sessions. Participants cycled for 90 min at 75 % $\dot{V}O_{2max}$ and then to exhaustion at 85 % $\dot{V}O_{2max}$ with this performance test being repeated on successive days after 2 weeks of normal training. Times-to-exhaustion were not different between treatments but performance during the final testing session was significantly reduced with carbohydrate supplementation. Ratings of muscle soreness before or after the performance tests were unaffected by supplementation although plasma CK levels were reduced with the combined carbohydrate + BCAA supplement after the third cycling test. Although the authors concluded addition of BCAA provided no advantage in terms of performance, it may have been more appropriate to compare the responses separately between tests 1 and 2 and then between tests 2 and 3 since the intervening amount of time (2 weeks vs. 1 day) and amount of training (2 weeks vs. none) between sessions was not constant. Nevertheless, these data further suggest there is little relationship between markers of muscle damage and exercise performance and provide no evidence supporting an immediate benefit of carbohydrate + protein supplementation before, during and after regular training sessions over an extended 2-week period.

Building upon their earlier findings of a delayed ergogenic benefit of carbohydrate + protein supplementation on sprint interval cycling performance [58], Thomson et al.

Table 3 A summary of the studies that followed changes in muscle damage, soreness and recovery of muscle function for many days or weeks with repeated bouts of cycling, running, marching or resistance exercise together with protein and/or carbohydrate supplementation

Study	Type of exercise	Supplementation dose and timing	Muscle damage (CK, LDH, Mb)	Muscle soreness	Muscle function
Cathcart et al. [61], TR 2 groups (23)	Cycle road race	53 g/h CHO 56 g/h CHO + 16 g/h PRO For 5–6 h during eight successive stages of road race	CHO + PRO = CHO	CHO + PRO = CHO	CHO + PRO < CHO for time for eight stages
Flakoll et al. [56], UT 3 groups (387)	Basic recruit training	8 g/day CHO 8 g/day CHO + 10 g/day PRO PLA After exercise every other day for 54 days		PRO < [CHO = PLA]	PRO < [CHO = PLA] for visits to medical clinic
Nelson et al. [16], TR (12)	Sprint cycling	60 g CHO 44 g CHO + 13.8 g PRO Every 30 min for 3 h or 1 h recovery during 5 days of training	CHO + PRO < CHO		CHO + PRO = CHO for sprint power after 3 and 6 days
Sharp and Pearson [62], UT (8)	Resistance training	6 g/day PRO PLA Morning and evening equal doses for 4 weeks	PRO < PLA		PRO = PLA for TW
Skillen et al. [57], TR (12)	Cycle training	23 g CHO 18 g CHO + 5 g PRO 500 mL 3×/day	CHO + PRO < CHO	CHO + PRO = CHO	CHO + PRO = CHO for TTE CHO + PRO > CHO for vertical jump
Thomson et al. [59], TR (10)	Sprint cycling	1.6 g/kg/h CHO 1.2 g/kg/h CHO + 0.45 g/kg/h PRO For 1.5 h recovery on three successive training days	CHO + PRO < CHO	CHO + PRO = CHO	CHO + PRO > CHO for sprint power after 3 days

CHO carbohydrate, PLA placebo, PRO protein, TTE time-to-exhaustion, TW total work, $\dot{V}O_{2max}$ maximal aerobic power

^a Participant numbers are provided in parentheses

[59] examined effects of a leucine-protein + carbohydrate supplement provided during a 1.5-h recovery period that followed 2.5 h of high-intensity interval cycling on three successive days. Leucine has attracted considerable attention in recent years given its ability to independently stimulate muscle protein synthesis [60]. Daily total protein intake was maintained at 1.6 g·kg⁻¹, although nitrogen balance determinations revealed participants were in slightly negative nitrogen balance during both conditions. There was a small but significant increase in overall sprint power and a small decrease in plasma CK levels with the leucine-protein + carbohydrate supplement compared with isocaloric CHO. However, the overall change in CK was low and there was no difference between supplements for ratings of muscle soreness.

This same laboratory then used kinetic estimates of leucine and glucose turnover at rest, during and following

exercise to evaluate mechanisms potentially accounting for the apparent performance benefit of leucine-protein supplementation [16]. Exercise sessions were extended over a 6-day period with sprint interval performance being compared on days 4 and 6. Daily protein intake was increased from 1.5 to 1.9 g·kg⁻¹ for the leucine-protein + carbohydrate condition, which reflected the additional leucine and protein provided immediately following the exercise sessions. In contrast to their earlier findings, power output during the sprint intervals was not different between conditions on either day 4 or 6. Also, participants were in positive nitrogen balance while receiving both supplements. The increase in plasma CK following exercise on days 4 and 6 was reduced with leucine-protein supplementation. The authors suggested the ergogenic effects of protein supplements on subsequent exercise performance are only evident when participants are in negative daily

nitrogen balance. The authors acknowledged that, although increased rates of protein synthesis during recovery and exercise could lead to lower levels of muscle damage and reduce plasma CK levels, these differences did not affect subsequent exercise performance.

Effects of carbohydrate + protein supplementation over the eight stages of the TransAlp Challenge road race were examined by Cathcart et al. [61]. Cyclists were randomly assigned to receive ad libitum liquid and solid carbohydrate or carbohydrate + protein supplements during each stage of the race, which was performed in warm environmental conditions averaging 33 °C. Plasma CK and muscle soreness increased similarly for both groups compared with pre-race levels and remained elevated throughout the competition. The carbohydrate + protein group completed every stage of the race, except the first, faster than those who only received the carbohydrate supplement. Since consumption during each race stage was ad libitum, the carbohydrate + protein group consumed more calories each hour while racing, which likely contributed to improved performance.

3.3.3 A Study Using Resistance Training

To assess whether BCAA supplementation prevents muscle damage, Sharp and Pearson [62] provided a BCAA supplement for three weeks before an experimental week that included four resistance exercise sessions. During the experimental week, BCAA supplements were administered in the morning and evening. Consistent with earlier studies [52, 57], plasma CK levels were reduced during the week of training with BCAA supplementation. However, there was no difference in the total work performed during each exercise session, implying that changes in the markers of muscle damage resulting from BCAA supplementation did not translate into ability to generate greater force during training.

3.3.4 Summary: Multiple Days of Exercise and Supplementation

It is apparent, therefore, that as the period of exercise and supplementation is extended over several days and weeks, surrogate markers of muscle damage and ratings of muscle soreness are reduced when protein is provided before, during and/or after exercise (Table 3) [16, 56, 57, 59, 62]. Changes in performance metrics were occasionally found to be related to the reductions in muscle damage and soreness [56, 59] but several other studies reported no impact of an extended period of protein supplementation alone or in combination with carbohydrate on performance [16, 57, 62] or an increase in performance with no change in measures of muscle damage or soreness [61].

4 Discussion

4.1 General Study Limitations

Many of the studies reviewed in this systematic analysis used a between-group rather than a repeated-measures experimental design [42, 44, 53, 54], due to the adaptation effects that have been observed from a single bout of novel eccentric exercise on changes in plasma CK levels when additional eccentric exercise is performed several weeks later [63]. Studies that used a repeated-measures design typically recruited trained cyclists or experienced resistance-trained participants to lessen this adaptation effect [16, 35–37, 39–41, 55, 58, 59] or separated their repeated testing sessions by over 10 weeks for those unaccustomed to the performance test [47]. Numbers of participants tested were not always justified with calculations of statistical power [41, 42, 51, 53–55, 59, 61] and loss of data [35, 42, 61] or initial small sample sizes [41, 44, 50, 55] increased Type II error risk. There has also been concern about the value of effect size inference statistical analyses for guiding decisions about the real-world relevance of the study results, rather than relying on traditional null hypothesis testing [64, 65]. Those studies that included inferential statistical analyses [16, 36, 39, 51, 59] provided little support for a physiological relationship between changes in muscle damage and muscle function.

The control of dietary intake is critical for meaningful comparisons to be made across trials involving nutritional supplements. Some studies provided no indication that diet was controlled [33, 42] or indicated participants were instructed to maintain their normal diet but dietary records were not collected [44, 50, 53]. Studies that maintained strict dietary control by providing meals to participants found no relationship between the recovery of muscle function and markers of muscle damage in response to supplementation [16, 51, 54, 57, 59, 61].

It is also unclear whether a sensitivity analysis, similar to the one conducted by Amann [66] for cycling time-trial and tests-to-exhaustion, have been performed for measurements of muscle strength using isokinetic, isometric, one repetition maximum or total-work methodology. Since some of the investigations incorporated more than one method to evaluate changes in muscle strength for the same or different muscle groups [47, 48, 50, 51, 54], the sensitivity of these various test metrics may not be similar and could increase the likelihood of Type I error.

4.2 Muscle Damage, Soreness and Recovery of Muscle Function

Several studies recruited participants unaccustomed to cycling [43] or resistance exercise [51–54] to ensure robust changes in muscle soreness and damage occurred in

response to an acute exercise bout under placebo or control conditions. With such untrained individuals, ingestion of a protein supplement often [51–53], but not always [54], reduced surrogate markers of muscle damage. However, improvements in muscle function were not associated with reduced indices of muscle damage [51–53].

It is also questionable whether the findings of these studies apply to endurance- or resistance-trained individuals, since they are accustomed to regular exercise and unlikely to exhibit the same level of muscle damage following an acute bout of exercise as untrained subjects [49, 67–69]. The findings by Howarth et al. [19] suggest protein supplementation following endurance exercise in regularly active individuals increases rates of muscle protein synthesis and attenuates rates of muscle protein breakdown. Indirect measures of muscle damage also appear to increase far less when protein supplementation is provided following a single bout of cycling, running or resistance-exercise for regularly active participants [35–37, 39–41, 55, 58] or following repeated daily training sessions for cyclists [16, 59] compared with 8 days of 4–5 h·day⁻¹ of competitive cycling through the mountains [61] or following running or cycling protocols that involved novel eccentric work [33, 34, 44], although the changes are highly variable across participants. Thus protein supplements may be more successful at reducing muscle damage when the exercise itself, such as submaximal running or cycling, is less likely to disrupt contractile proteins.

Studies that evaluated the recovery of muscle function for less than a 24-h period typically used a time-trial or cycling test-to-exhaustion as their physical performance metric (see Table 1) and found there was no relationship between changes in muscle damage or soreness that followed this acute bout of endurance exercise and subsequent performance with protein supplementation [33–37, 39–41]. When the acute exercise involved eccentric loading, and performance metrics involved tests of muscle strength that were recorded over several days (see Table 2), there was also little evidence to support a relationship between changes in markers of muscle damage or soreness and recovery of muscle function with protein supplementation [43–45, 47, 48, 50–55]. However, when protein supplementation continued for extended periods (see Table 3), ratings of muscle soreness and indirect markers of muscle damage appear to have been reduced prior to and/or following repeated bouts of exercise [16, 56–59, 62]. However, once again these changes are not always reflected by changes in physical performance or muscle function [16, 57, 62], although there was some indication that a relationship did exist [56, 57, 59]. Importantly, as stated by Nelson et al. [16], and later confirmed by Lunn et al. [15], the ergogenic effects associated with protein supplementation on physical performance during or following an acute exercise bout or over several days of training may only be evident if participants are in negative nitrogen balance without the use of supplements.

As reviewed by others [6], the use of surrogate markers of muscle injury, such as changes in plasma CK, are poor indicators of the damage to contractile proteins following an acute bout of exercise and would not be expected to relate to subsequent changes in muscle function. In addition, reliance on qualitative measures of muscle soreness as the sole indicator of muscle damage should be discouraged in favor of using more reliable quantitative measures of muscle function, including isokinetic measures of peak torque and maximal voluntary contraction, and joint range of motion [70]. Nevertheless, studies of the effects of protein supplementation on recovery of muscle function rely on surrogate markers of muscle damage and subjective measures of soreness and this represents a major limitation with these studies. Instead, research is needed that relates direct measures of myofibrillar disruption to measures of muscle function [71–73]. In addition, use of stable isotope labeling and direct measure of intracellular protein signaling and gene expression [15, 38] suggest protein supplementation creates a more anabolic state post-exercise within muscle, which may reduce extent of muscle damage associated with an acute bout of exercise.

5 Evidence Statement—Ingestion of Protein to Reduce Muscle Damage and Soreness and Improve Muscle Function

There are consistent, and high quality experimental data demonstrating that regardless of whether a protein supplement is consumed alone or together with carbohydrate prior to, during or following an acute bout of endurance or resistance exercise there is little relationship between recovery of muscle function, ratings of muscle soreness and surrogate markers of muscle damage. *Evidence category A.* There are limited, quality experimental data to demonstrate that ingestion of a protein supplement before, during or following an acute bout of exercise or after daily training sessions will attenuate muscle soreness and/or reduce markers of muscle damage. *Evidence category B.* To date there are limited, quality experimental data to demonstrate that ingestion of a protein supplement before, during or following an acute bout of exercise will confer an ergogenic advantage during subsequent exercise if participants are in negative nitrogen balance without supplement use. *Evidence category B.*

6 Future Research

The intent of this manuscript was to provide a systematic evaluation of the literature that examined the hypothesis that acute and repeat consumption of supplemental protein

promotes recovery of muscle function and performance in response to resistance- and endurance-type exercise by attenuating muscle damage and soreness. It is evident, that surrogate markers of muscle damage and subjective measures of muscle soreness are not reliable indicators of the biological effects of protein supplementation. When combined with the inherent limitations of applied muscle and physical performance tests, it is not surprising that literature does not strongly support the hypothesis that protein supplementation enhances recovery by attenuating muscle damage. However, the acute benefits of supplemental protein on skeletal muscle protein balance (synthesis > breakdown) cannot be ignored [12, 74–77], as the stimulation of protein synthesis with concomitant reductions in protein breakdown are critical to remodel skeletal muscle.

It is hoped that this review will provide the impetus for new research designed to elucidate the mechanisms by which enhanced protein balance affects skeletal muscle and whole-body physical performance. Furthermore, considering that the digestive properties and amino acid profile (particularly leucine) of supplemental protein may differentially impact skeletal muscle protein synthesis [78] and long-term protein accretion [79], studies that determine whether manipulating the source (whey vs. casein vs. a combined milk protein supplement) and timing of protein supplementation during acute and chronic muscle recovery should be conducted. For example, whey protein is readily digested, rapidly increasing amino acid availability, thereby making it ideal to consume post exercise. However, casein is a slowly digested protein that elicits a lower postprandial rise in amino acid levels and muscle protein synthesis, consequently it may be more appropriate to consume before sleep to maximize overnight recovery from exercise [80]. Other research might target the development of new approaches to assess applied muscle and physical function that align more closely with the acute and long-term consequence of supplemental protein on muscle protein synthesis. Studies should also combine applied outcomes with kinetic and molecular assessments of muscle biology (e.g., protein turnover, intracellular signaling, and miRNA) to identify targets that predispose beneficial skeletal muscle outcomes to expand the work by Drummond et al. [81, 82] and Davidsen et al. [83]. Collectively, findings from these studies would provide the necessary evidence-base to support or refute the hypothesis that protein supplementation promotes recovery by attenuating muscle damage.

7 Concluding Remarks

This review assessed the existing evidence-base that supports use of protein supplements before, during or after an

acute bout of exercise to reduce indices of muscle damage and enhance recovery of muscle function. Limitations in study designs, which included small sample numbers and the lack of dietary control, together with the large variability in surrogate markers of muscle damage reduced the strength of the evidence-base. The purported mechanism that relates protein supplementation with reduced muscle damage following exercise and, therefore, a faster recovery of muscle function does not appear to be supported by the current literature. However, a growing body of evidence has focused on the benefits of protein supplements on post-exercise muscle anabolism, which, in theory, could affect the recovery of muscle function. Clearly, additional research with supplementation is essential to determine if there is a causal link between the time course for the changes in post-exercise protein synthesis, associated anabolic intracellular signaling, and recovery of muscle function.

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